

Question did not ask for explanation. So no need to explain.

9(a)	Radioactive decay is a spontaneous and random process whereby an unstable nucleus disintegrates into a more stable configuration by the emission of radiation e.g. alpha-particles, beta-particles or gamma rays. Nuclear fission is the breaking up of a large nucleus into 2 or more smaller nuclei of comparable mass, with the emission of a few neutrons and/or other radiations. Nuclear fusion is the formation of a larger nucleus from 2 small nuclei, with the possible emission of other radiations.
9(b)(i)	Using $E = mc^2$ $= 14.003074 + 4.002604 - 16.999130 - 1.007825 \times (1.66 \times 10^{-27}) \times (3 \times 10^8)^2$ $= 1.9078 \times 10^{-13}$ $\approx 1.91 \times 10^{-13} \text{ J}$
9(b)(ii)	The minimum energy required for the reaction to take place is $1.91 \times 10^{-13} \text{ J}$. Since kinetic energy of $1.76 \times 10^{-13} \text{ J}$ is less than that, the reaction will not take place.
9(c)(i)	Using $E = hf = h \frac{c}{\lambda}$, $\Rightarrow \lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{1.16 \times 10^6 \times 1.6 \times 10^{-19}}$ $= 1.0717 \times 10^{-12}$ $\approx 1.07 \times 10^{-12} \text{ m}$

No.	Solution
9(c)(ii)	Using de Broglie's relation, $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{1.0717 \times 10^{-12}}$ $= 6.1866 \times 10^{-22}$ $\approx 6.19 \times 10^{-22} \text{ Ns}$
9(d)(i)	By conservation of momentum, $p_{\text{nucleus}} + p_{\gamma} = 0$ $\therefore p_{\text{nucleus}} = 6.1866 \times 10^{-22}$ $E_k = \frac{1}{2}mv^2$ $= \frac{m^2v^2}{2m}$ $= \frac{p^2}{2m}$ $= \frac{(6.1866 \times 10^{-22})^2}{2 \times 60 \times 1.66 \times 10^{-27}}$ $= 1.9214 \times 10^{-18}$ $\approx 1.92 \times 10^{-18} \text{ J}$
9(d)(ii)	The energy of $1.92 \times 10^{-18} \text{ J}$ ($= 1.2 \times 10^{-5} \text{ MeV}$) is small compared to energy of γ - ray photon ($= 1.16 \text{ MeV}$)